

Experiment No. 2
Apply data cleaning techniques on the given dataset
Date of Performance: 03/02/2025
Date of Submission: 24/02/2025



Department of Computer Engineering

Academic Year: 2024-25

Class / Branch: BE Computer

Semester: VIII

Subject: Applied Data Science Lab

Experiment No. 2

Aim: Apply data cleaning techniques by cleaning the given dataset by handling missing values, duplicates, inconsistencies, outliers, and standardizing formats for better data quality

Dataset: In this experiment, a fictitious data containing 10 observations and 4 variables is used. The dataset contains Age, Salary, and joining date, salary and department columns. The dataset has categorical variables and missing values in these columns.

Software used: Google Colaboratory / Jupyter Notebook

Theory :-

Data cleaning is just the collective name to a series of actions we perform on our data in the process of getting it ready for analysis.

Some of the steps in data cleaning are:

- Handling missing values
- Encoding categorical features
- Outliers detection
- Transformations etc.

Handling missing values is a key part of data preprocessing and hence, it is of utmost importance for data scientists/machine learning engineers to learn different techniques in relation imputing / replacing numerical or categorical missing values with appropriate value based on appropriate strategies. That's primarily the reason we need to convert categorical columns to numerical columns so that a machine learning algorithm understands it. This process is called categorical encoding.

SimpleImputer is a class found in package sklearn.impute. It is used to impute / replace the numerical or categorical missing data related to one or more features with appropriate values.

Typically, any structured dataset includes multiple columns – a combination of numerical as well as categorical variables. A machine can only understand the numbers. It cannot understand the text. That's essentially the case with Machine Learning algorithms too. There are multiple ways of handling Categorical variables.

The two most widely used techniques:

- Label Encoding
- One-Hot Encoding

Label Encoding is a popular encoding technique for handling categorical variables. In this technique, each label is assigned a unique integer based on alphabetical ordering.

One-Hot Encoding is the process of creating dummy variables. It simply creates additional features based on the number of unique values in the categorical feature. Every unique value in the category will be added as a feature.

Program

<https://colab.research.google.com/drive/18yMPuwvfC0hHSQ8bp5Skt3Ncwgu5e7gO>

```

Data after handling missing values:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11 entries, 0 to 10
Data columns (total 7 columns):
 #   Column          Non-Null Count  Dtype
---  -
 0   ID              11 non-null    float64
 1   Name            11 non-null    object
 2   Age             11 non-null    float64
 3   Gender          11 non-null    object
 4   Salary          11 non-null    float64
 5   Joining_Date    11 non-null    object
 6   Department      11 non-null    object
dtypes: float64(3), object(4)
memory usage: 748.0+ bytes
None

```

	ID	Name	Age	Gender	Salary	Joining_Date	Department
0	1.0	Alice	25.0	Female	50000.0	2021-06-01	HR
1	2.0	Bob	30.0	Male	60000.0	2020-07-15	IT
2	3.0	Charlie	29.2	M	70000.0	2019/08/20	HR
3	4.0	David	40.0	Male	61700.0	2022-05-30	Finance
4	5.0	Eve	22.0	Female	55000.0	01-01-2018	IT
5	6.0	Frank	35.0	F	72000.0	2017-03-22	IT
6	6.0	Frank	35.0	Male	72000.0	2017-03-22	IT
7	7.0	Grace	29.0	F	49000.0	2017-03-22	HR
8	8.0	Henry	50.0	Male	80000.0	2016-12-11	Finance
9	9.0	Frank	31.0	Unknown	57000.0	2023-02-05	Unknown
10	10.0	Ivy	-5.0	Female	52000.0	2015-09-27	HR

Data after handling outliers:

```
<class 'pandas.core.frame.DataFrame'>
```

Index: 9 entries, 0 to 9

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	ID	9 non-null	float64
1	Name	9 non-null	object
2	Age	9 non-null	float64
3	Gender	9 non-null	object
4	Salary	9 non-null	float64
5	Joining_Date	9 non-null	object
6	Department	9 non-null	object

dtypes: float64(3), object(4)

memory usage: 576.0+ bytes

None

	ID	Name	Age	Gender	Salary	Joining_Date	Department
0	1.0	Alice	25.0	Female	50000.0	2021-06-01	HR
1	2.0	Bob	30.0	Male	60000.0	2020-07-15	IT
2	3.0	Charlie	29.2	M	70000.0	2019/08/20	HR
3	4.0	David	40.0	Male	61700.0	2022-05-30	Finance
4	5.0	Eve	22.0	Female	55000.0	01-01-2018	IT
5	6.0	Frank	35.0	F	72000.0	2017-03-22	IT
6	6.0	Frank	35.0	Male	72000.0	2017-03-22	IT
7	7.0	Grace	29.0	F	49000.0	2017-03-22	HR
9	9.0	Frank	31.0	Unknown	57000.0	2023-02-05	Unknown

Final cleaned data:

```
<class 'pandas.core.frame.DataFrame'>
```

Index: 9 entries, 0 to 9

Data columns (total 7 columns):

#	Column	Non-Null Count	Dtype
0	ID	9 non-null	float64
1	Name	9 non-null	int64
2	Age	9 non-null	float64
3	Gender	9 non-null	int64
4	Salary	9 non-null	float64
5	Joining_Date	9 non-null	int64
6	Department	9 non-null	int64

dtypes: float64(3), int64(4)

memory usage: 576.0 bytes

None

	ID	Name	Age	Gender	Salary	Joining_Date	Department
0	1.0	0	25.0	1	50000.0	4	1
1	2.0	1	30.0	3	60000.0	3	2
2	3.0	2	29.2	2	70000.0	2	1
3	4.0	3	40.0	3	61700.0	5	0
4	5.0	4	22.0	1	55000.0	0	2
5	6.0	5	35.0	0	72000.0	1	2
6	6.0	5	35.0	3	72000.0	1	2
7	7.0	6	29.0	0	49000.0	1	1
9	9.0	5	31.0	4	57000.0	6	3

Conclusion :-

The experiment successfully demonstrated various data cleaning techniques, including handling missing values, encoding categorical variables, and detecting outliers. By applying these methods, the dataset was transformed into a more structured and analyzable format. The use of tools like SimpleImputer helped in efficiently managing missing data, ensuring better data quality. Encoding techniques such as Label Encoding and One-Hot Encoding enabled seamless conversion of categorical variables for machine learning models. Overall, the experiment highlighted the importance of preprocessing in data science. Future improvements can focus on automating data cleaning for larger datasets.